

LendWise: SBA Approved Loan Amount Prediction

AI Final Project Report

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Course: CS5100 / CSE 587B Final Project

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Dataset: SBA National Loan Dataset

Target: GrAppv, the gross approved loan amount

Best model: LightGBM | RMSE \$184,928 | MAE \$98,792 | R2 0.572

1. Introduction

Problem Statement

This project predicts the approved dollar amount for small business loans using the SBA National Loan Dataset. The target variable is GrAppv, the gross amount approved by the lender. The project is a regression task.

Motivation and Goals

Small business lending decisions depend on loan term, business size, state, lender location, industry, program flags, and whether the business is new or existing. A model that estimates an expected approved amount can support realistic expectations and closer review of unusual cases.

2. Background

The SBA lending data contains historical records across banks, states, industries, and years. Approved amount is highly skewed, so the task is harder than a simple linear prediction problem.

The project uses supervised regression. Linear models provide simple baselines, while tree-based ensemble models learn nonlinear relationships and interactions between features.

3. Methodology

Tools and Libraries

- Python, Jupyter Notebook, pandas, NumPy, SciPy
- scikit-learn for preprocessing, models, PCA, and metrics
- LightGBM, XGBoost, and CatBoost for boosted tree regression
- TensorFlow/Keras with GPU support for the neural network
- Matplotlib and Seaborn for EDA and diagnostics

Dataset and Scope

The raw SBA file contains 899,164 rows and 27 columns before setting the loan number as the index. After target cleanup, 899,164 usable rows remain.

The model excludes direct leakage and post-approval performance fields such as SBA_Appv, DisbursementGross, MIS_Status, ChgOffPrinGr, and charge-off dates.

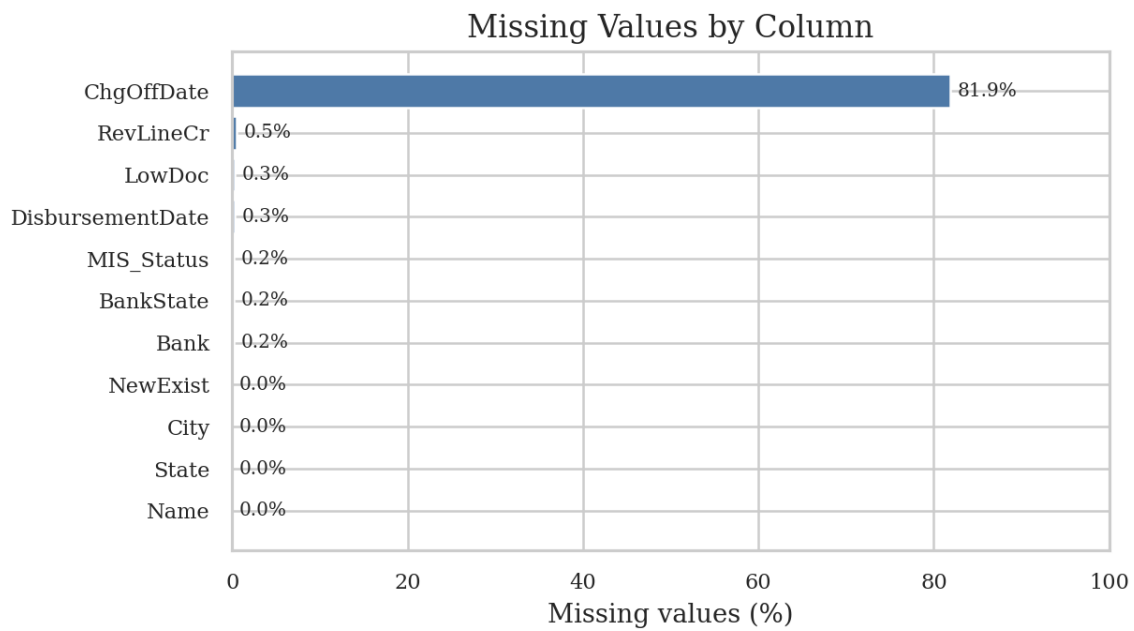


Figure 1. Missing-value profile of the SBA dataset.

Preprocessing and Feature Engineering

- 1 Cleaned currency columns and date fields.
- 2 Created approval year, month, quarter, and decade features.
- 3 Created term, employment, job-ratio, industry, geography, and lender features.
- 4 Filled numeric missing values with medians and categorical missing values with Unknown.
- 5 One-hot encoded categorical variables using compact int8 dummy columns and safe feature names.

The final modeling table uses 19 numeric features and 14 categorical features before encoding. After one-hot encoding, the model input contains 381 features.

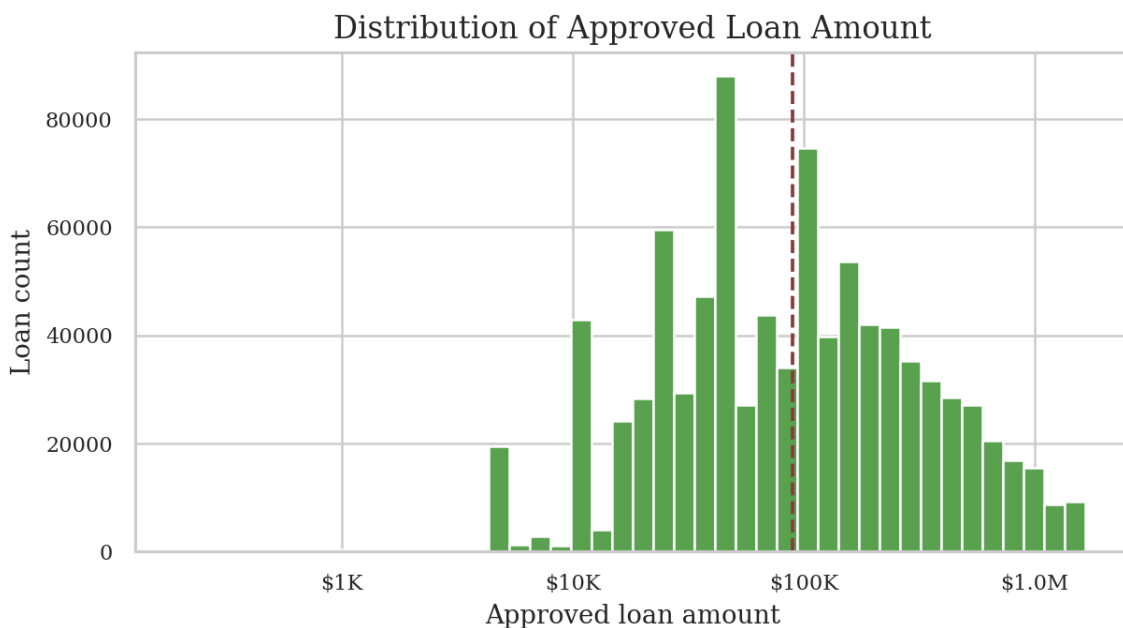


Figure 2. Approved loan amount distribution.

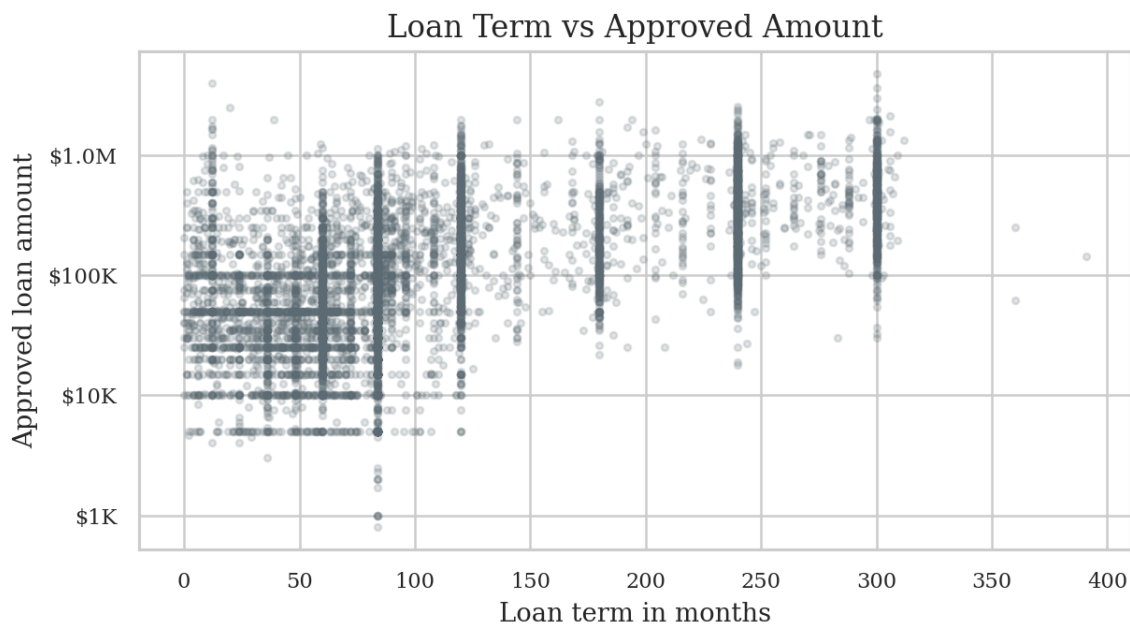


Figure 3. Loan term versus approved amount on a log dollar scale.

Training and Evaluation Design

The dataset was split into 575,464 training rows, 143,867 validation rows, and 179,833 test rows. Models are evaluated with MAE, RMSE, R2, and RMSLE.

4. Implementation

The implementation is contained in `src/LendWise.ipynb`. The notebook downloads the dataset when missing, cleans data, performs EDA, builds the modeling table, trains models, and generates evaluation plots.

The final model list is Linear Regression, Ridge Regression, PCA + Linear Regression, Gradient Boosting, Random Forest, HistGradientBoosting, LightGBM, XGBoost, CatBoost, and Neural Network. TensorFlow detected the NVIDIA GeForce RTX 4060 Laptop GPU in the final run.

5. Results and Evaluation

The best model is LightGBM, with RMSE \$184,928, MAE \$98,792, R2 0.572, and RMSLE 0.951. The top models are close, which suggests that boosted/tree-based methods are a good fit for the engineered SBA features.

Rank	Model	MAE	RMSE	R2	RMSLE
1	LightGBM	\$98,792	\$184,928	0.572	0.951
2	XGBoost	\$99,469	\$186,165	0.566	0.869
3	HistGradientBoosting	\$99,842	\$186,498	0.565	0.946
4	Random Forest	\$98,301	\$186,855	0.563	0.788
5	CatBoost	\$102,672	\$190,672	0.545	0.956

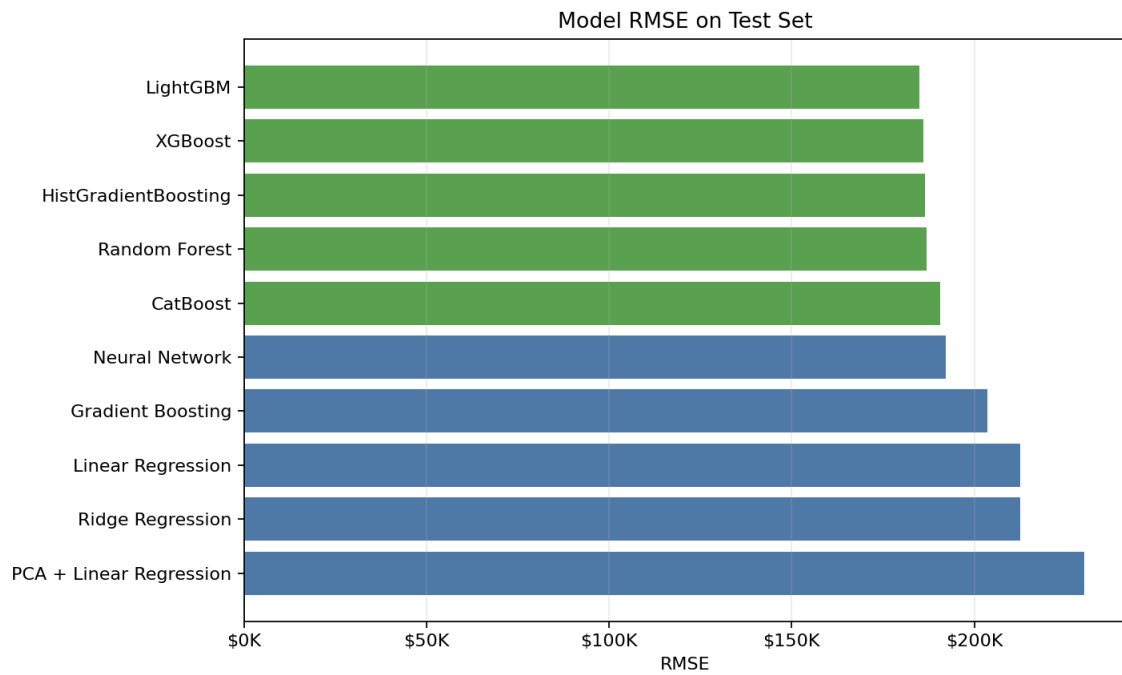


Figure 4. Test RMSE by model. Lower is better.

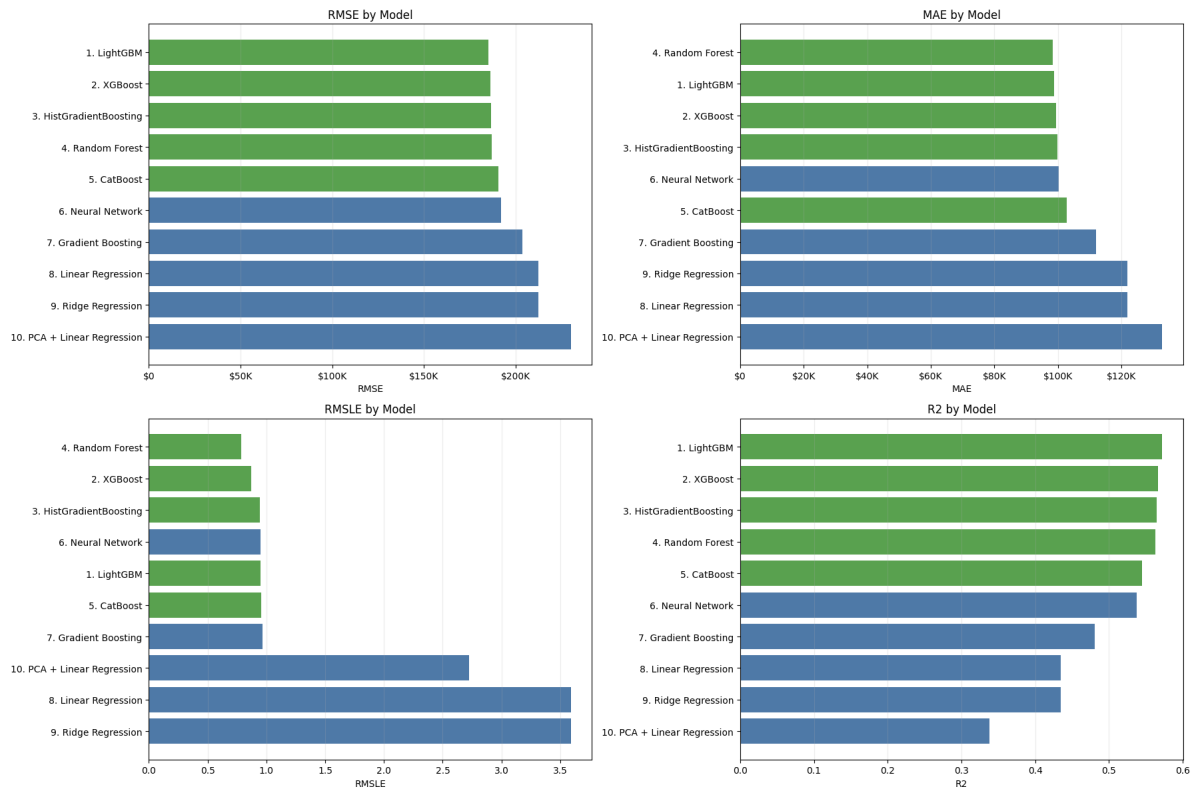


Figure 5. Model comparison across RMSE, MAE, RMSLE, and R2.

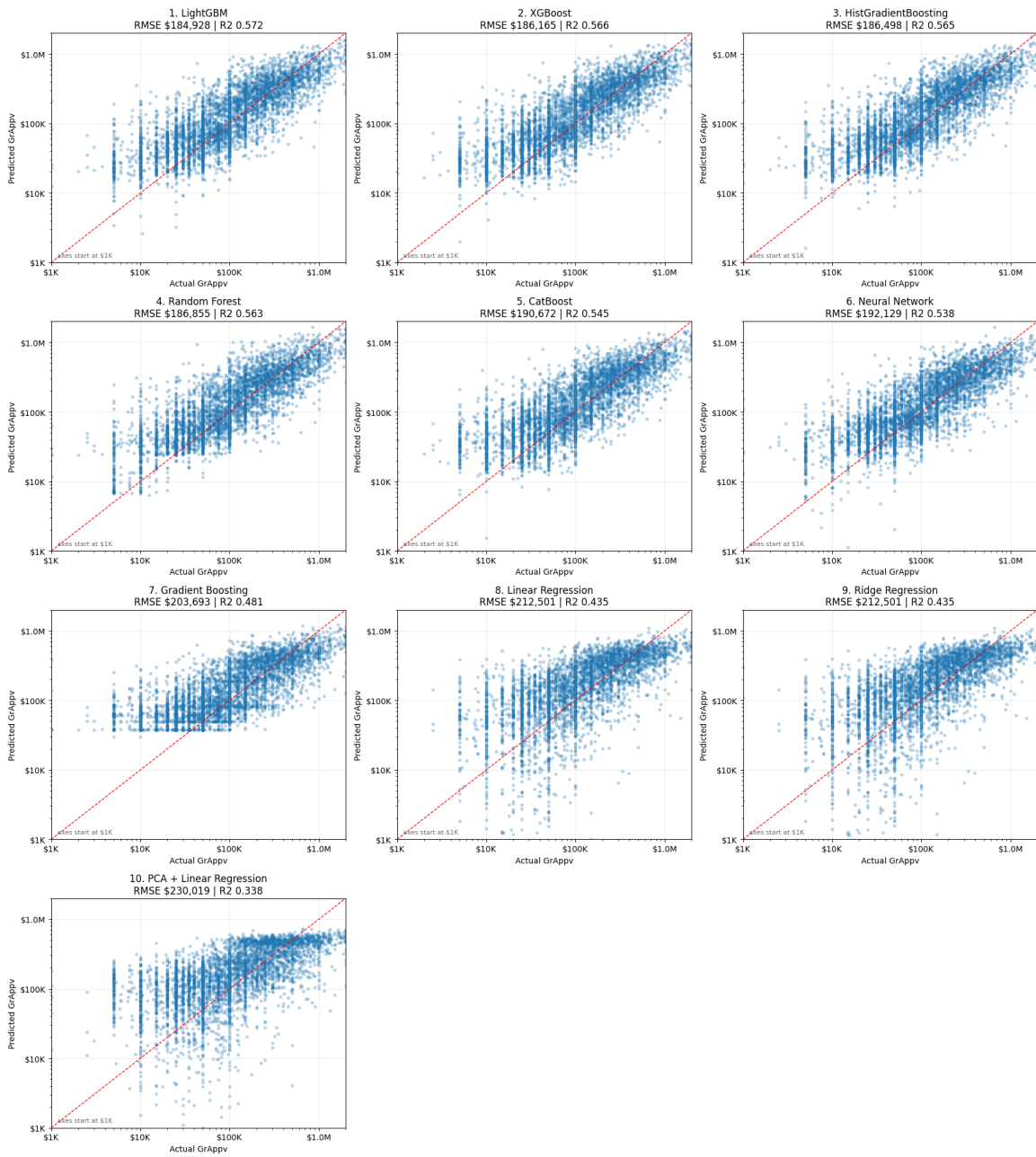


Figure 6. Predicted versus actual approved amount. Axes start at \$1K and use a log scale.

Validation by Loan Size

The overall RMSE hides important differences across loan sizes. The test set was split into small loans below \$50K, medium loans from \$50K to \$250K, and large loans above \$250K.

Loan Size	Rows	MAE	RMSE	R2	Mean Error
Small (<\$50K)	69933	\$35,805	\$59,072	-14.673	\$32,033
Medium (\$50K-\$250K)	70098	\$83,430	\$125,453	-4.256	\$47,534
Large (>\$250K)	39802	\$236,515	\$347,369	0.135	\$-140,322

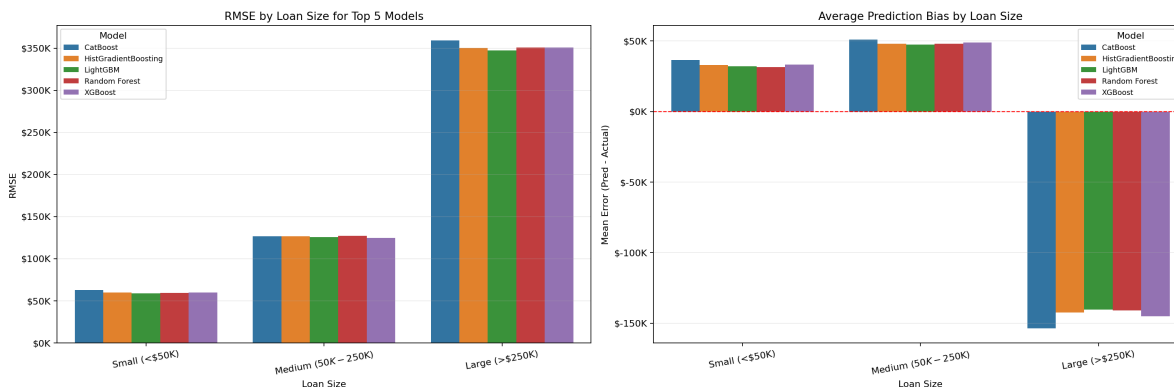


Figure 7. RMSE and average prediction bias by loan size for the top 5 models.

The loan-size validation shows that LightGBM over-predicts small and medium loans on average, but under-predicts large loans. This confirms the regression-toward-the-middle pattern seen in the scatter plots.

6. Discussion

Limitations

- The target is highly skewed, so large loans strongly affect RMSE.
- Important underwriting variables such as revenue, collateral, credit profile, and requested amount are missing.
- A time-based split may be more realistic than a random split.
- Predictions are less reliable at the smallest and largest loan amounts.

Potential Improvements

- Tune LightGBM, XGBoost, and HistGradientBoosting.
- Try time-based validation.
- Train separate models by loan-size segment or add a two-stage model that first predicts loan-size group.
- Use native categorical handling in CatBoost and LightGBM.
- Build a small prediction interface for demonstration.

7. Demonstration and Setup Instructions

- 1 Create and activate the project virtual environment.
- 2 Install dependencies from requirements.txt.
- 3 Open src/LendWise.ipynb in Jupyter.
- 4 Run the dataset download cell if src/data/SBAnational.csv is missing.
- 5 Run all cells to reproduce preprocessing, model training, evaluation tables, and plots.

8. Conclusion

The project successfully converts LendWise into an SBA approved loan amount regression system. The preprocessing now matches SBA lending data, leakage columns are removed, engineered features are added, and multiple models are evaluated fairly on the same test set.

LightGBM is the final recommended model because it has the lowest RMSE and highest R2 among the tested models. The loan-size validation shows that exact prediction remains hardest for very large approvals.

Individual Contribution

Deepak completed the dataset change to SBA National, preprocessing redesign, feature engineering, model training, GPU setup, evaluation updates, plot diagnostics, loan-size validation, and report preparation. This section can be edited if additional team members need to be listed.

9. References

- SBA National Loan Dataset. Kaggle. <https://www.kaggle.com/datasets/maddraf/sbanational-csv>
- U.S. Small Business Administration. 7(a) Loan Program overview. <https://www.sba.gov/funding-programs/loans/7a-loans>
- scikit-learn documentation. <https://scikit-learn.org/stable/>
- LightGBM documentation. <https://lightgbm.readthedocs.io/>
- XGBoost documentation. <https://xgboost.readthedocs.io/>
- CatBoost documentation. <https://catboost.ai/>
- TensorFlow documentation. <https://www.tensorflow.org/>

Appendix: Split-Level Loan-Size Accuracy

Regression does not use classification accuracy. This appendix reports practical accuracy-style measures for LightGBM, the best overall model.

Split	Loan Size	Rows	MAE	RMSE	R2	Tol. Acc.	25% Acc.
Train	Small (<\$50K)	223397	\$35,757	\$58,275	-14.318	55.5%	18.1%
Train	Medium (\$50K-\$250K)	224933	\$82,708	\$123,567	-4.096	48.1%	29.1%
Train	Large (>\$250K)	127134	\$233,496	\$343,366	0.167	33.3%	37.1%
Validation	Small (<\$50K)	56006	\$35,855	\$58,847	-14.480	55.5%	18.3%
Validation	Medium (\$50K-\$250K)	56094	\$83,606	\$125,396	-4.237	47.6%	28.9%
Validation	Large (>\$250K)	31767	\$236,906	\$349,393	0.122	33.1%	36.7%
Test	Small (<\$50K)	69933	\$35,805	\$59,072	-14.673	55.3%	18.1%
Test	Medium (\$50K-\$250K)	70098	\$83,430	\$125,453	-4.256	48.0%	29.0%
Test	Large (>\$250K)	39802	\$236,515	\$347,369	0.135	33.1%	36.4%

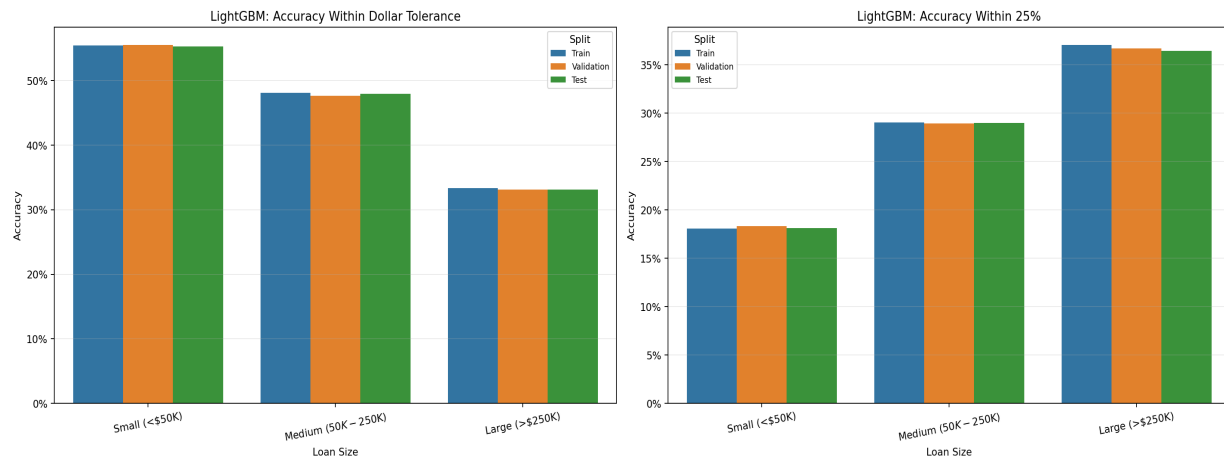


Figure 8. LightGBM train, validation, and test accuracy by loan size.

The train, validation, and test values are very close. This means the main problem is not overfitting. Instead, the model is biased toward middle loan amounts and lacks enough high-signal borrower information to make precise predictions at the extremes.